

Network Devices

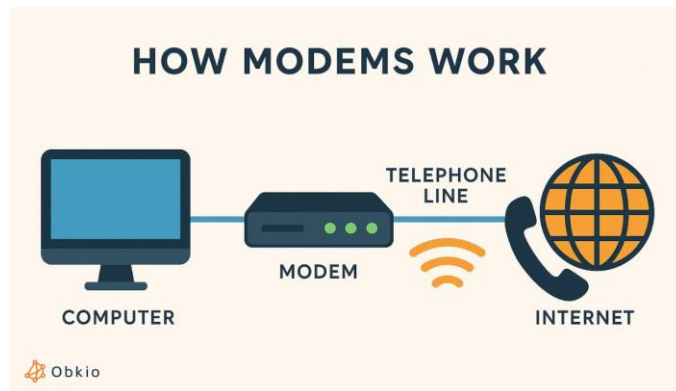
Network devices are hardware components used to **connect computers and other devices, transmit data, and facilitate communication within and between networks.**

MODEM

Modem = Modulator + Demodulator

Key Points

- A modem enables a computer/network to communicate over certain communication lines by converting signals.
- **Modulation:** Converts digital data into a suitable signal for transmission.
- **Demodulation:** Converts received signals back into digital data.
- Traditionally used with telephone-based communication systems.
- A modem may connect a local network to an Internet service.



Ethernet Card

An Ethernet card is also known as a **Network Interface Card (NIC)** when it provides wired network connectivity.

Key Points

- Provides a computer with a **wired network interface**.
- Allows a computer to communicate with other devices over an Ethernet network.
- Usually provides an **RJ45 Ethernet port**.
- Has a unique **MAC address** used for network communication.
- May be built into the computer's motherboard or installed separately.

Remember:

Ethernet Card → Wired network connectivity

RJ45 Connector

RJ45 is commonly used to refer to the **8P8C modular connector** used with Ethernet network cables.

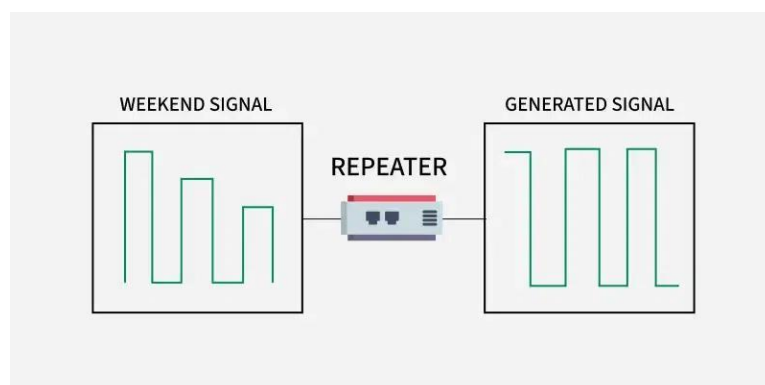
Key Points

- Used for connecting Ethernet cables to networking equipment.
- Commonly found on **Ethernet cables and network ports**.
- Frequently associated with **twisted-pair Ethernet cabling**.
- It is a **connector**, not a networking device that processes or routes data.
- Used to establish a physical wired connection.

Remember:

RJ45 → Connector used for Ethernet connections

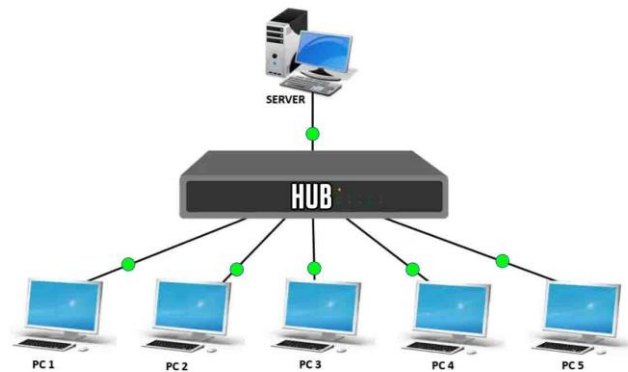
Repeater



Key Points

- A repeater is used to **regenerate or retransmit signals**.
- Helps extend the transmission distance of a network.
- Receives a weakened signal and regenerates it before sending it onward.
- Operates at the **Physical Layer (Layer 1)** of the OSI model.
- Does not normally make decisions about the destination of packets.

HUB



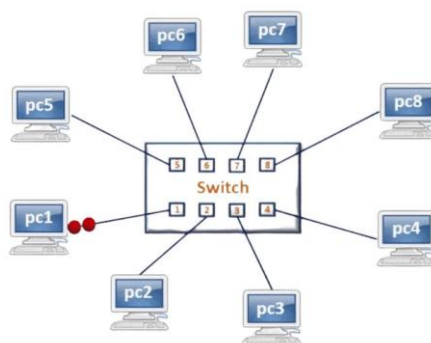
Key Points

- A hub connects multiple devices in a network.
- It receives data through one port and **broadcasts it to all other ports**.
- Does not intelligently determine the intended recipient.
- Generally operates at the **Physical Layer (Layer 1)**.
- Less efficient than a switch because unnecessary traffic is generated.

Remember:

Hub → Broadcasts data to all connected ports

SWITCH



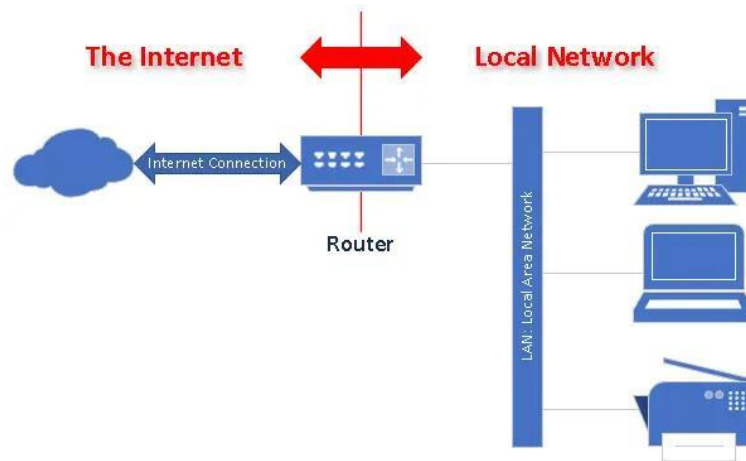
Key Points

- A switch connects multiple devices within a LAN.
- Forwards data to the **appropriate destination port**.
- Uses **MAC addresses** to make forwarding decisions.
- More efficient than a hub.
- Generally operates at the **Data Link Layer (Layer 2)**.
- Commonly used in school, office, and enterprise LANs.

Remember:

Switch → Uses MAC address to forward data

ROUTER



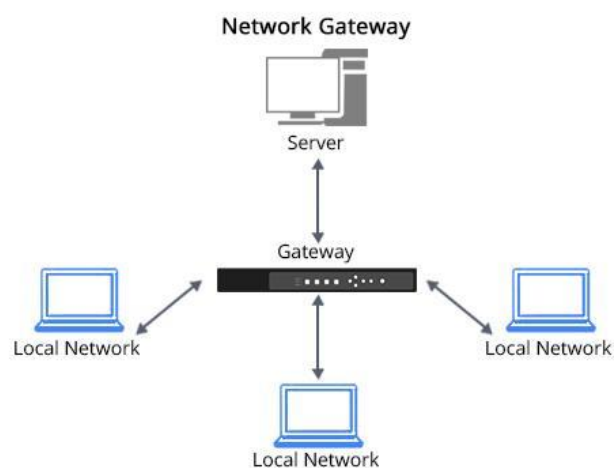
Key Points

- A router connects **different networks**.
- Determines an appropriate path for forwarding packets.
- Uses **IP addresses** for routing decisions.
- Commonly connects a LAN to another network or the Internet.
- Operates primarily at the **Network Layer (Layer 3)**.
- Maintains/uses routing information to determine where packets should go.

Remember:

Router → Connects networks and forwards packets using IP addresses

GATEWAY



Key Points

- A gateway acts as an **entry and exit point between different networks**.
- Can connect networks using **different protocols or communication architectures**.
- May perform protocol conversion when required.
- It can operate at different layers depending on its implementation.
- In many practical networks, the term **default gateway** refers to the device through which hosts send traffic destined for other networks.

Remember:

Gateway → Connects dissimilar networks/protocols

WiFi CARD



A Wi-Fi card is a **wireless network interface** that allows a computer or device to connect to a Wi-Fi network.

Key Points

- Provides **wireless network connectivity**.
- Uses radio waves to communicate with wireless access points/routers.
- May be built into a computer or available as an external adapter.
- Allows devices to join a **wireless LAN (WLAN)**.
- Has a network interface and associated MAC address.

Remember:

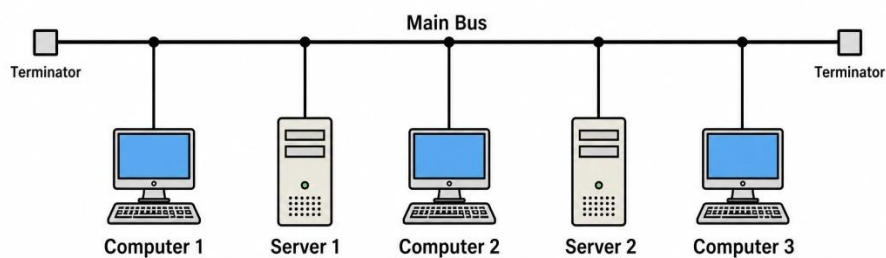
Wi-Fi Card → Wireless network connectivity

TOPOLOGY

1. BUS TOPOLOGY

Key Points

- All computers/devices are connected to a **single common cable** called the **backbone** or bus.
- Data travels along the shared backbone.
- **Terminators** are placed at both ends of the backbone.
- It requires comparatively **less cable**.
- Simple and economical for small networks.
- If the **main backbone fails, the entire network/segment may fail**.
- As the number of devices increases, network traffic can increase.
- Fault identification can be difficult.



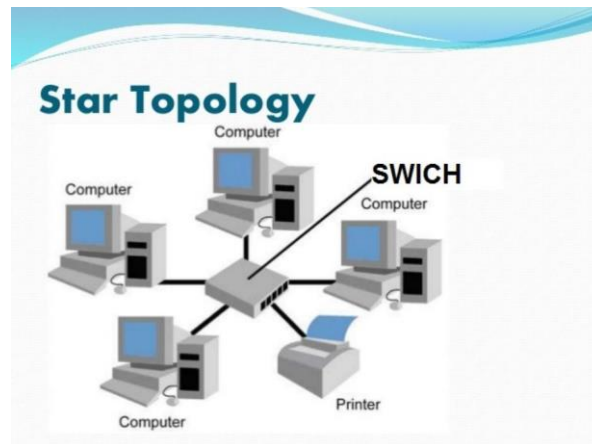
★ 2. STAR TOPOLOGY

Key Points

- Each computer/device is connected to a **central networking device**.
- The central device is generally a **hub or switch**.
- Each device has an independent connection to the central device.
- Easy to **install, manage and troubleshoot**.
- Failure of one individual cable generally affects only the connected device.
- Failure of the **central hub/switch can affect the entire network**.
- Commonly used for **LANs**.

Exam Point

Star = All devices connected to a central hub/switch



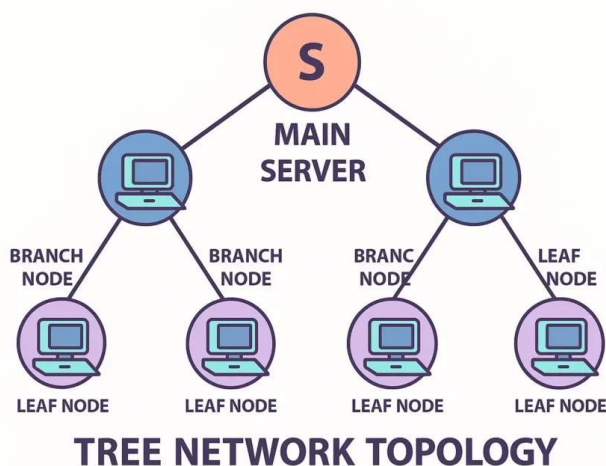
♣ 3. TREE TOPOLOGY

Key Points

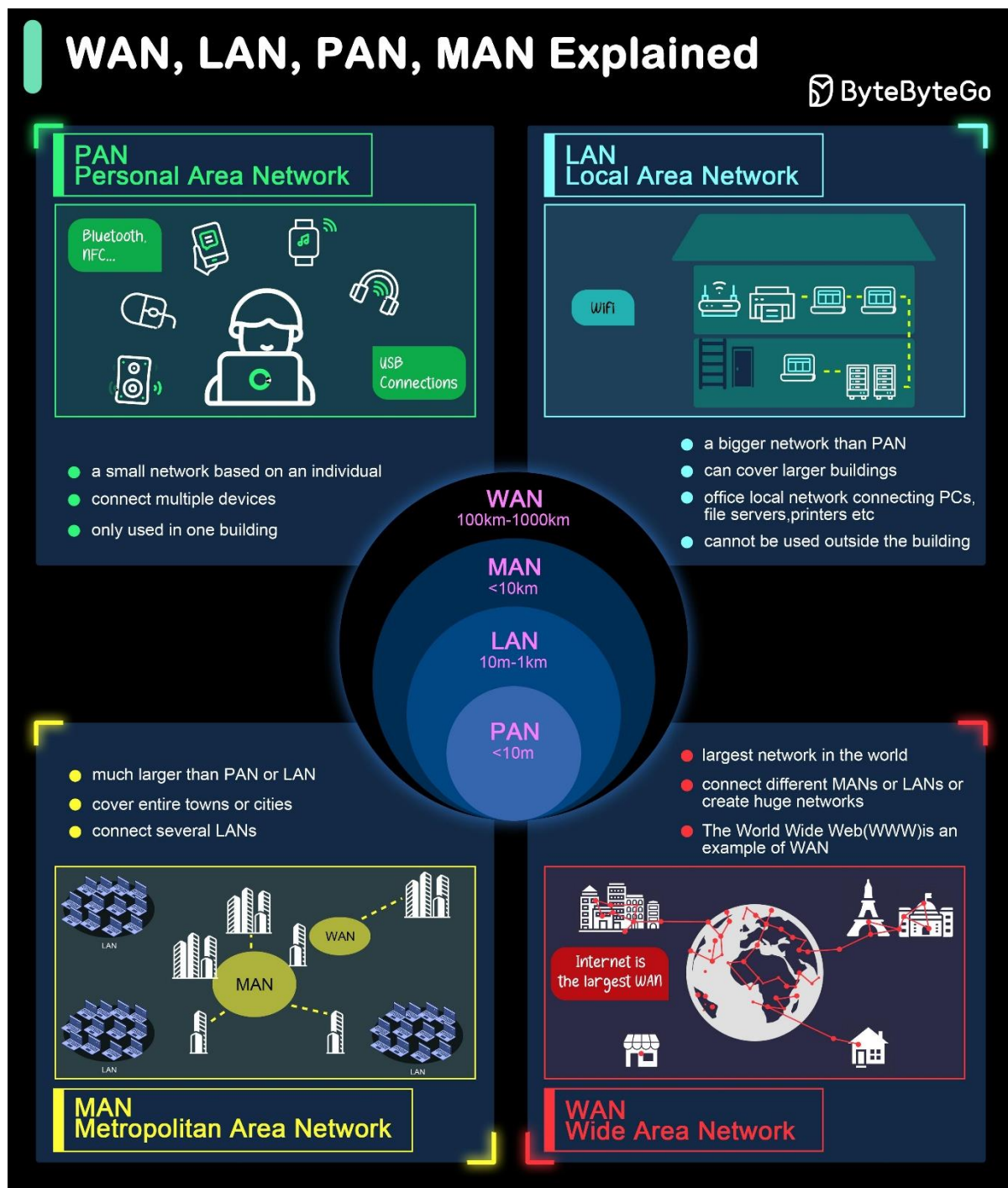
- Tree topology has a **hierarchical structure**.
- It consists of a **root/central node and multiple branches**.
- It can be viewed as a combination of **star and bus characteristics**.
- Lower-level devices are connected through intermediate networking devices.
- Suitable for **large and hierarchical networks**.
- It is relatively easy to expand by adding branches.
- Failure of an upper-level link/device can affect the corresponding branch.
- Requires more cabling and networking hardware than a simple bus topology.

📌 Exam Point

Tree = Hierarchical network with branches



TYPES OF NETWORKS



1. PAN – Personal Area Network

Key Points

- **PAN = Personal Area Network**
- Smallest type of network in this classification.
- Connects **personal devices** belonging to an individual.
- Approximate range: **up to 10 metres**.
- Can be **wired or wireless**.
- Bluetooth and USB connections are common examples.
- Examples:
 - Smartphone ↔ Smartwatch
 - Smartphone ↔ Bluetooth earbuds
 - Laptop ↔ Smartphone



Remember

PAN → Personal → Small area → About 10 m

2. LAN – Local Area Network

Key Points

- **LAN = Local Area Network**
- Connects devices within a **limited geographical area**.
- Can cover a room, floor, office, school, laboratory, or campus.
- NCERT describes LANs as extending up to approximately **1 km**.
- Provides **high-speed data transfer**.
- Can use Ethernet cables, fibre-optic cables or Wi-Fi.
- Usually privately managed.
- Examples:
 - School computer lab
 - Office network
 - Home Wi-Fi network
 - College campus network

Remember

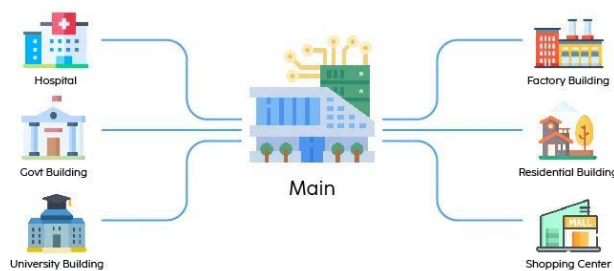
LAN → Local → Building/Campus → High speed

3. MAN – Metropolitan Area Network

Key Points

- **MAN = Metropolitan Area Network**
- Covers a **larger geographical area than LAN**.
- Generally covers a **city or town**.
- NCERT describes MAN as an extended form of LAN and notes that it can extend approximately **30–40 km**.
- Can connect several LANs.
- Data transfer rate is generally lower than that of a LAN.
- Examples:
 - City-wide cable TV network
 - Cable-based broadband network
 - Network connecting offices across a city

Metropolitan Area Network (MAN)



○

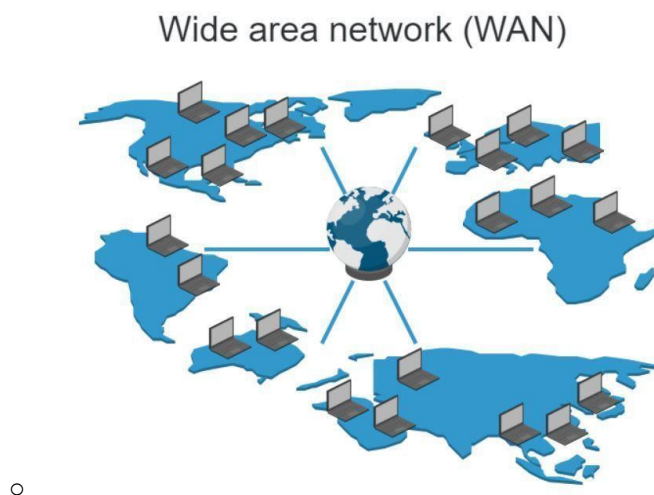
Remember

MAN → Metropolitan → City/Town → Connects LANs

4. WAN – Wide Area Network

Key Points

- **WAN = Wide Area Network**
- Covers a **very large geographical area**.
- Can connect networks across:
 - Different cities
 - Different states
 - Different countries
 - Different continents
- Connects **LANs and MANs** over large distances.
- Can use wired and wireless communication media.
- Generally more complex and expensive than LAN.
- **The Internet is the world's largest WAN.**
- Examples:
 - Banking networks connecting branches across countries
 - Multinational company networks



★ CBSE Must-Remember

PAN → Personal devices → ~10 m

LAN → Room/Building/School/Campus → up to ~1 km

MAN → City/Town → ~30–40 km

WAN → Countries/Continents → Internet